

# PROJECT REPORT

## Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau

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|---------------|------------------------------------------------------------------------------------|
| Date          | 06 July 2026                                                                       |
| Name          | Aditya Kumar Singh                                                                 |
| Project Name  | Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites in Tableau |
| Maximum Marks | —                                                                                  |

# 1. INTRODUCTION

## 1.1 Project Overview

The UNESCO World Heritage Sites Analysis (“Heritage Treasures”) is a data analytics and visualization project that explores the distribution, characteristics, and preservation status of the world's 1,121 UNESCO World Heritage Sites across 168 countries and five geographic regions. Raw, publicly available UNESCO heritage data was collected, cleaned, and modeled, then transformed into an interactive Tableau dashboard and guided Story. The finished analysis was further wrapped in a custom Flask web application (“UNESCO Heritage”) providing a Home, About, Dashboard, and Storyboard experience so that non-technical visitors can explore the findings without needing Tableau expertise.

## 1.2 Purpose

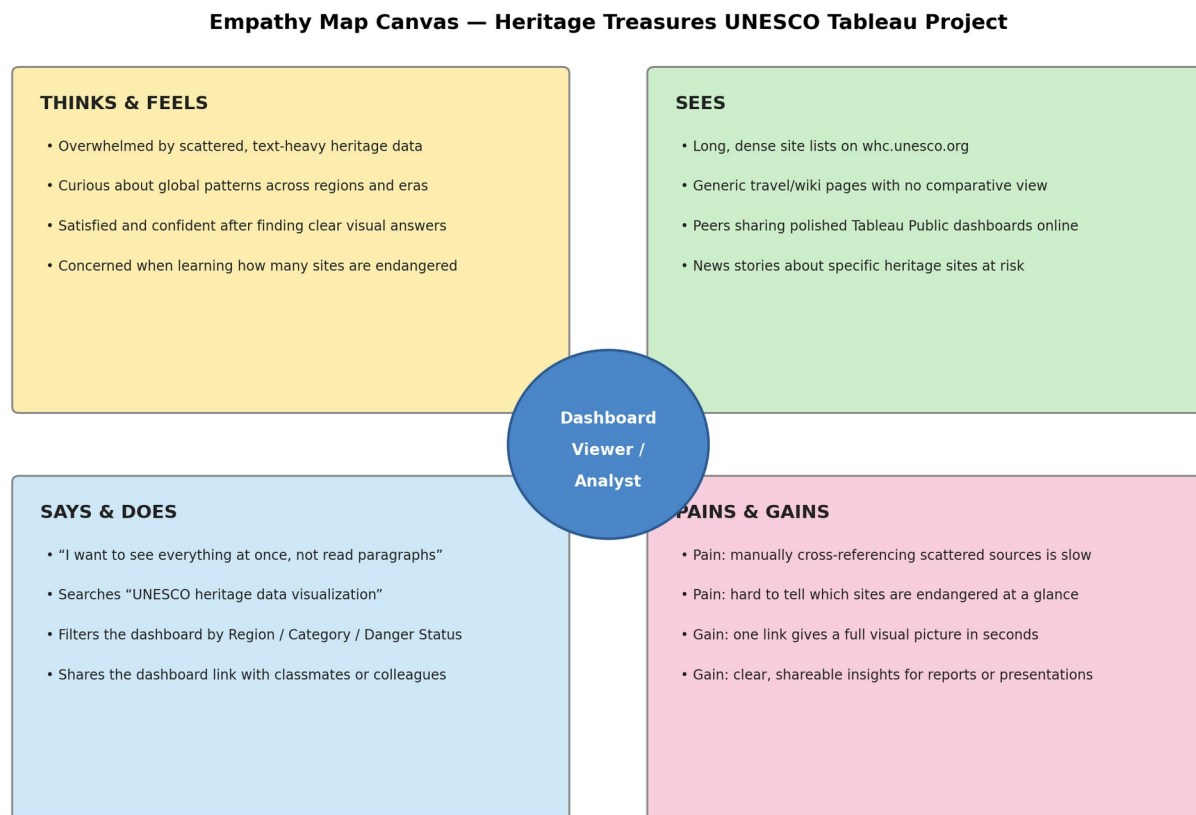
- Consolidate scattered, text-heavy UNESCO heritage data into a single, visual, explorable resource.
- Reveal patterns in cultural and natural heritage across regions, categories, and time that are not visible in raw spreadsheets.
- Raise awareness of endangered heritage sites (53 currently classified as “In Danger”) and the urgency of conservation.
- Provide students, researchers, educators, and travel enthusiasts with a free, self-serve tool for exploring global heritage trends.
- Demonstrate an end-to-end data analytics workflow: data collection, cleaning, visualization, and deployment.

## 2. IDEATION PHASE

### 2.1 Problem Statement

UNESCO World Heritage Site information is scattered across lengthy, text-heavy sources (the official site list, country pages, PDF reports). There is no easy, visual way for students, researchers, educators, or travelers to explore trends across all 1,121 sites by region, category, danger status, or year of inscription — making it hard to build a comparative, big-picture understanding of global heritage preservation.

### 2.2 Empathy Map Canvas



*Figure: Empathy Map Canvas for the Heritage Treasures dashboard viewer / analyst persona*

### 2.3 Brainstorming

The team brainstormed ideas across four themes — data sources, visualization approaches, delivery/presentation, and target audience — before converging on the final solution.

## Brainstorming — Heritage Treasures UNESCO Tableau Project

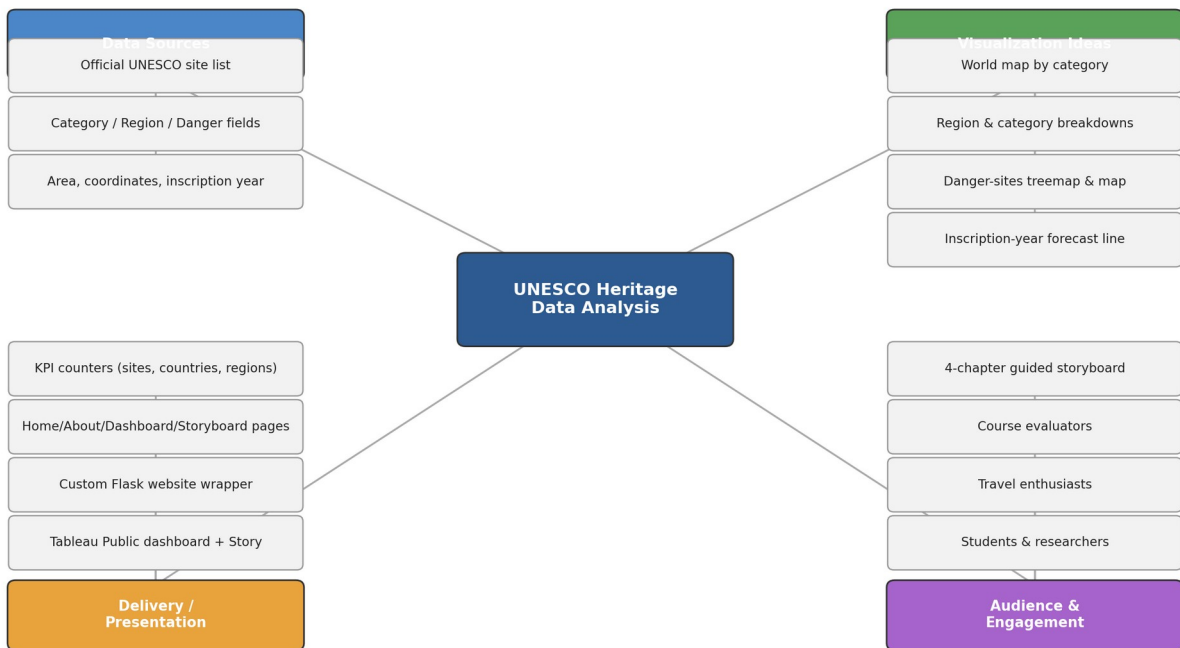


Figure: Brainstorming mind map for the Heritage Treasures project

## 3. REQUIREMENT ANALYSIS

### 3.1 Customer Journey Map

The customer journey was mapped across five stages — Entice, Enter, Engage, Exit, and Extend — covering how a visitor discovers, explores, and shares insights from the dashboard. Key steps include discovering the dashboard online, filtering by Region/Category/Danger Status, following the guided Story, and sharing the link with others. Full detail is provided in the accompanying Customer Journey Map document (submitted separately).

### 3.2 Solution Requirement

Functional requirements: user/viewer access, data collection & cleaning, data visualization, dashboard & story development, filtering & interactivity, and publishing & sharing. Non-functional requirements: usability, security, reliability, performance, availability, and scalability. Full detail is provided in the Solution Requirements document (submitted separately).

### 3.3 Data Flow Diagram

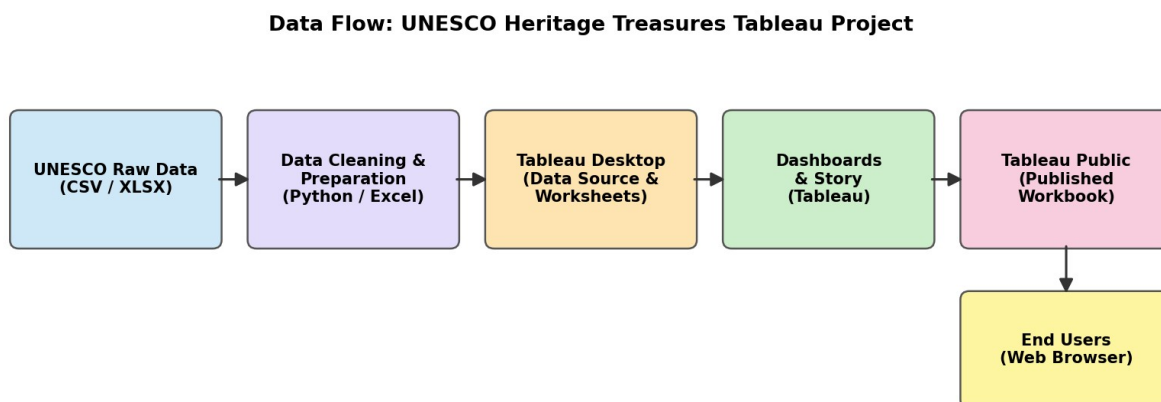
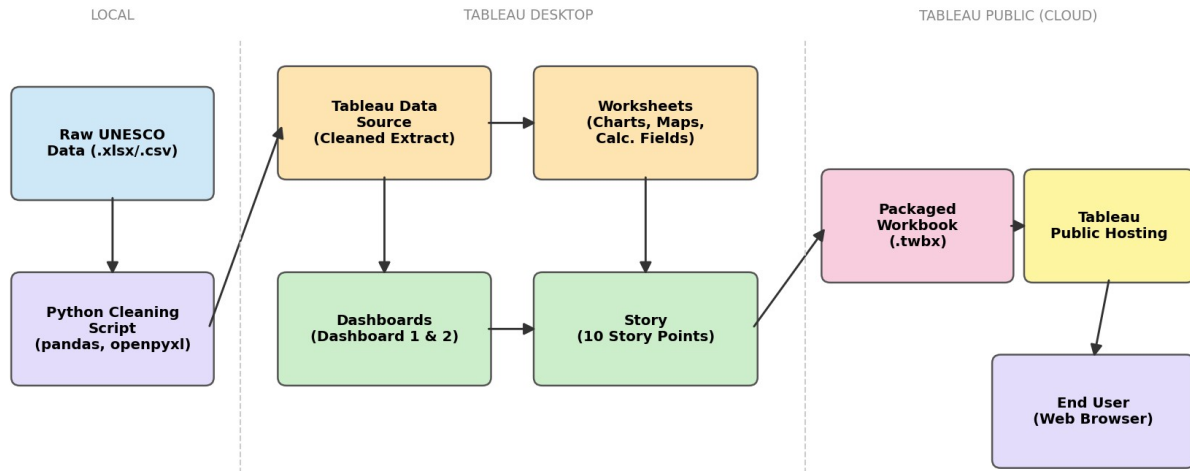


Figure: DFD Level 0 — data flow from raw UNESCO dataset through cleaning, Tableau visualization, and publishing to the dashboard viewer

### 3.4 Technology Stack



*Figure: Technology stack — Excel data source, Python cleaning, Tableau Desktop, Tableau Public hosting, and browser-based viewing*

In addition to the Tableau Public deployment, the dashboards were embedded in a custom Flask web application (Python, HTML/CSS, Jinja templates) providing Home, About, Dashboard, and Storyboard pages with KPI summaries and a four-chapter narrative.

## 4. PROJECT DESIGN

### 4.1 Problem Solution Fit

The Problem-Solution Fit canvas maps the target customer segments (students, researchers, travel enthusiasts, evaluators), their jobs-to-be-done (finding a consolidated, visual view of heritage data), root cause (no unified interactive UNESCO visualization exists), and the proposed solution — the Heritage Treasures Tableau dashboard and story. Full detail is provided in the Problem – Solution Fit document (submitted separately).

### 4.2 Proposed Solution

“Heritage Treasures: An In-Depth Analysis of UNESCO World Heritage Sites” — an interactive Tableau dashboard and story built on a fully cleaned dataset of all 1,121 sites, combining a world map, region/category breakdowns, a danger-sites analysis, and an inscription-year forecast, published as a single shareable Tableau Public link and presented through a custom Flask website. Full detail is provided in the Proposed Solution document (submitted separately).

### 4.3 Solution Architecture

**Solution Architecture — Heritage Treasures: UNESCO World Heritage Sites in Tableau**

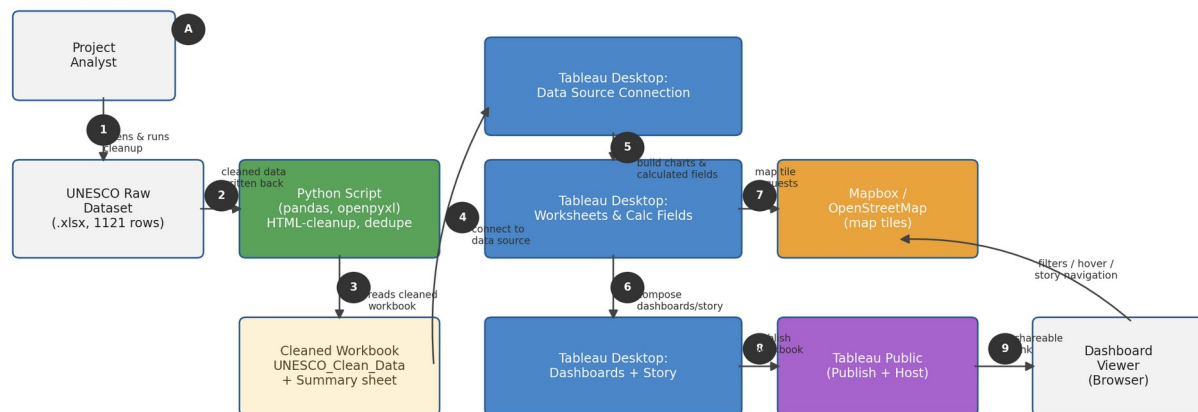
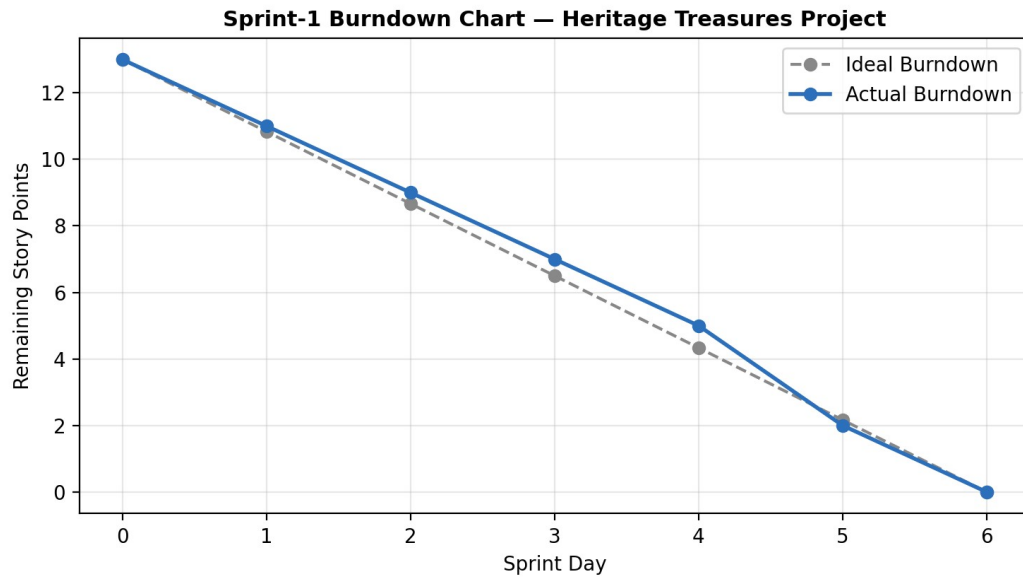


Figure: Solution architecture — from raw dataset and cleaning script through Tableau Desktop development to Tableau Public hosting and the dashboard viewer

## 5. PROJECT PLANNING & SCHEDULING

### 5.1 Project Planning

The project was delivered across two sprints. Sprint-1 (13 story points) covered data collection, cleaning, and core visualization development. Sprint-2 (10 story points) covered dashboard assembly, story development, publishing, and documentation. Both sprints completed on their planned end dates with no carry-over work.



*Figure: Sprint-1 burndown chart — ideal vs. actual story points remaining*

Full backlog, sprint schedule, velocity calculations, and burndown analysis are provided in the Project Planning Template document (submitted separately).



## **6. FUNCTIONAL AND PERFORMANCE TESTING**

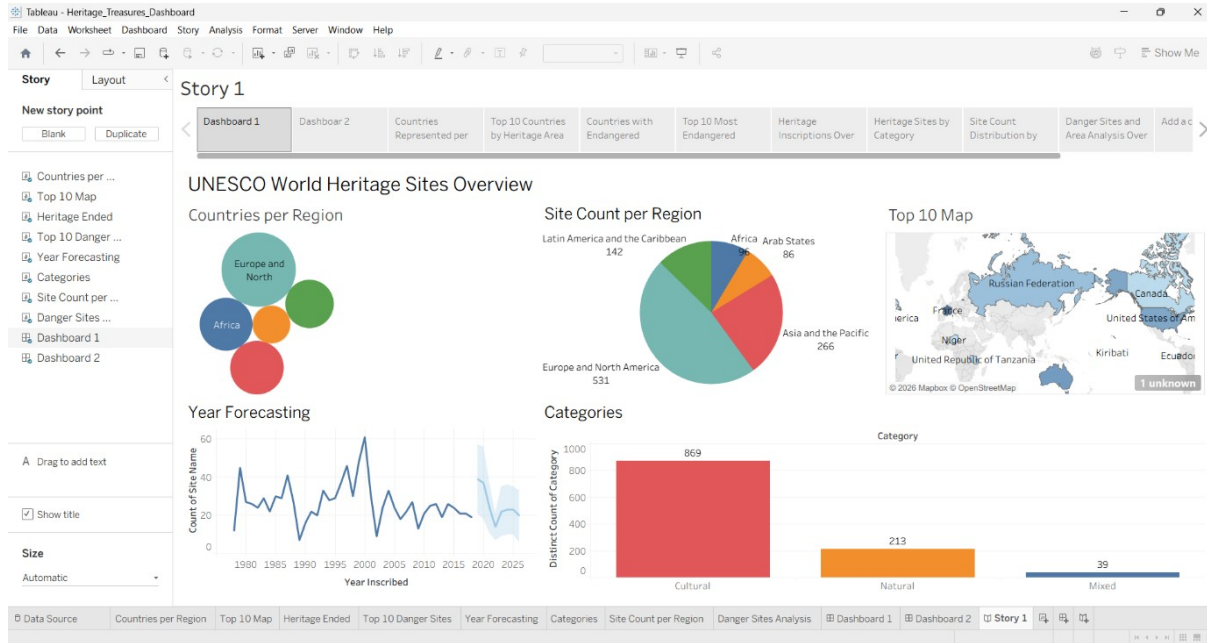
### **6.1 Performance Testing**

Model performance testing covered: data rendered (1,121 sites across 18 fields from a single Excel extract), data preprocessing (HTML-entity/whitespace cleanup, duplicate verification), utilization of filters (Region, Category, Danger Status), calculation fields used (Distinct Count of Site Name, SUM(Area (Hectares)), Forecast Indicator), dashboard design (7 visualizations across 2 dashboards), and story design (9 story points in the guided narrative). Full detail with screenshots is provided in the Project Performance Test document (submitted separately).

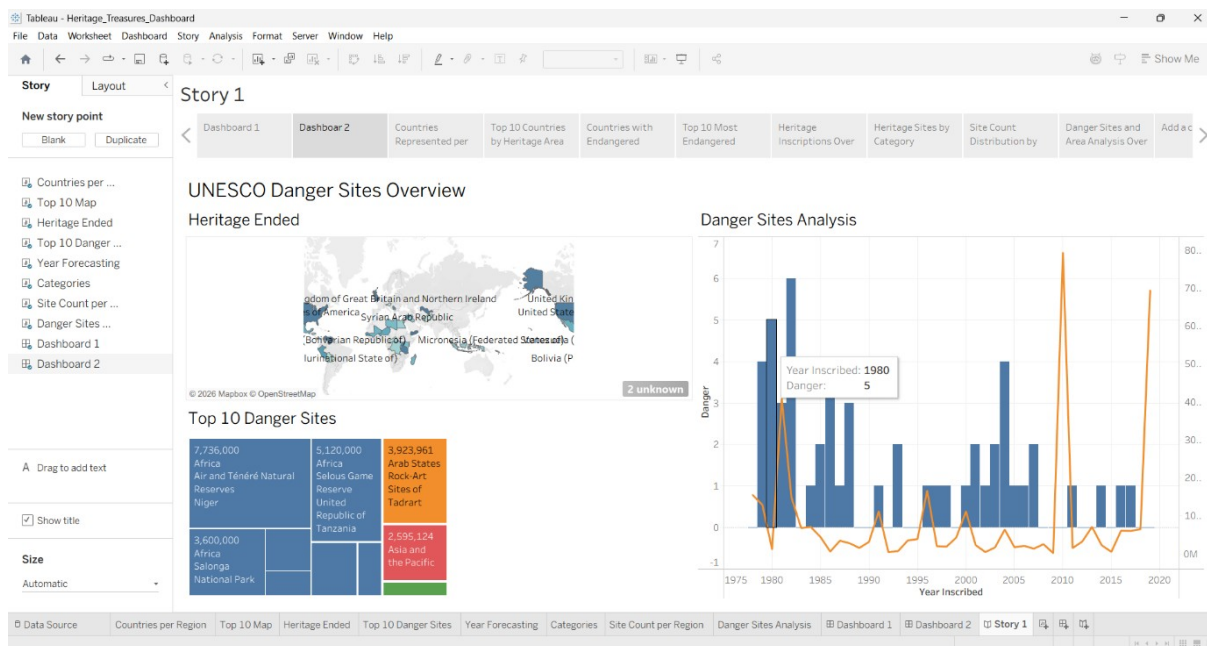
## 7. RESULTS

### 7.1 Output Screenshots

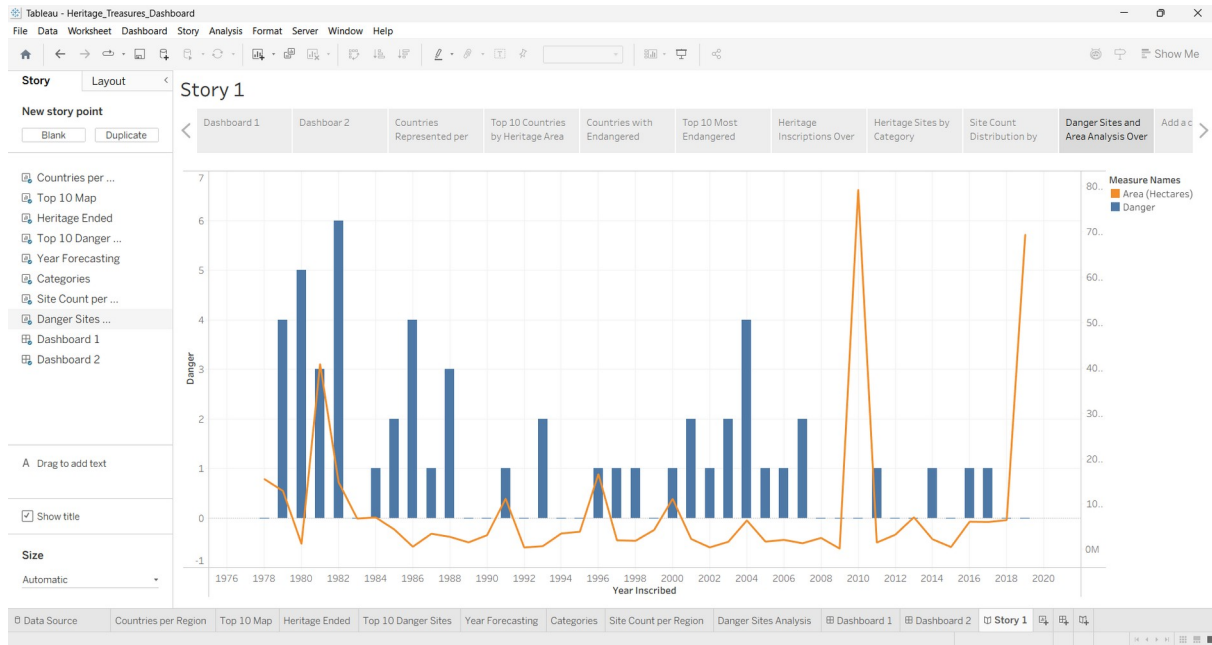
#### Tableau Dashboards and Story:



#### Dashboard 1 — UNESCO World Heritage Sites Overview



#### Dashboard 2 — UNESCO Danger Sites Overview

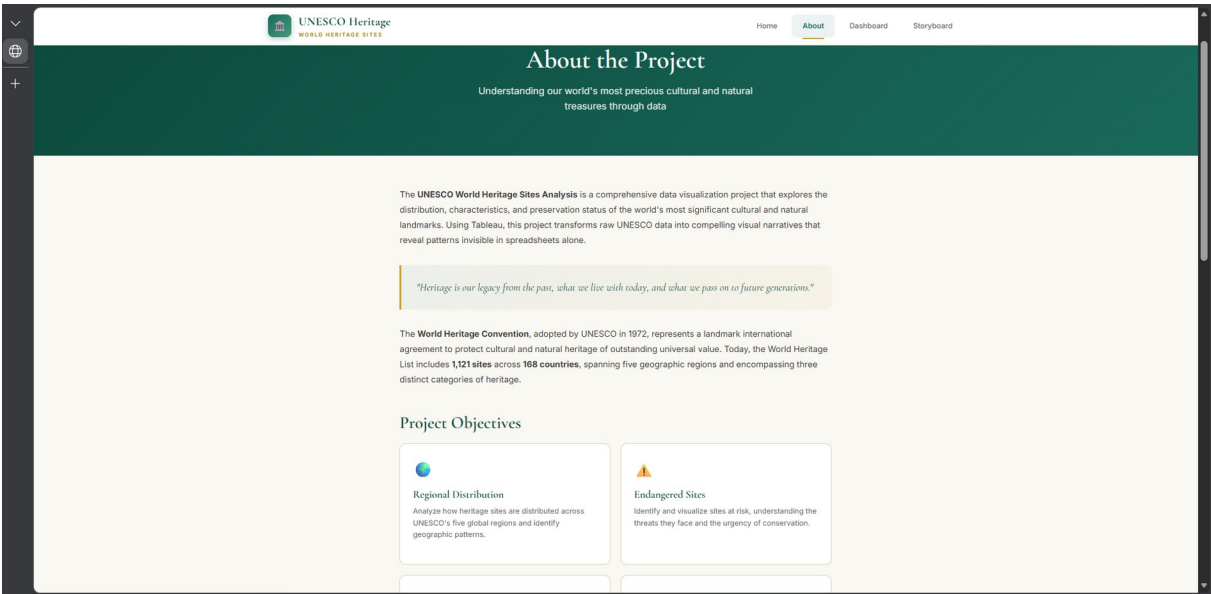


*Story 1 — Danger Sites and Area Analysis (final story point)*

**Flask Web Application (project presentation website):**



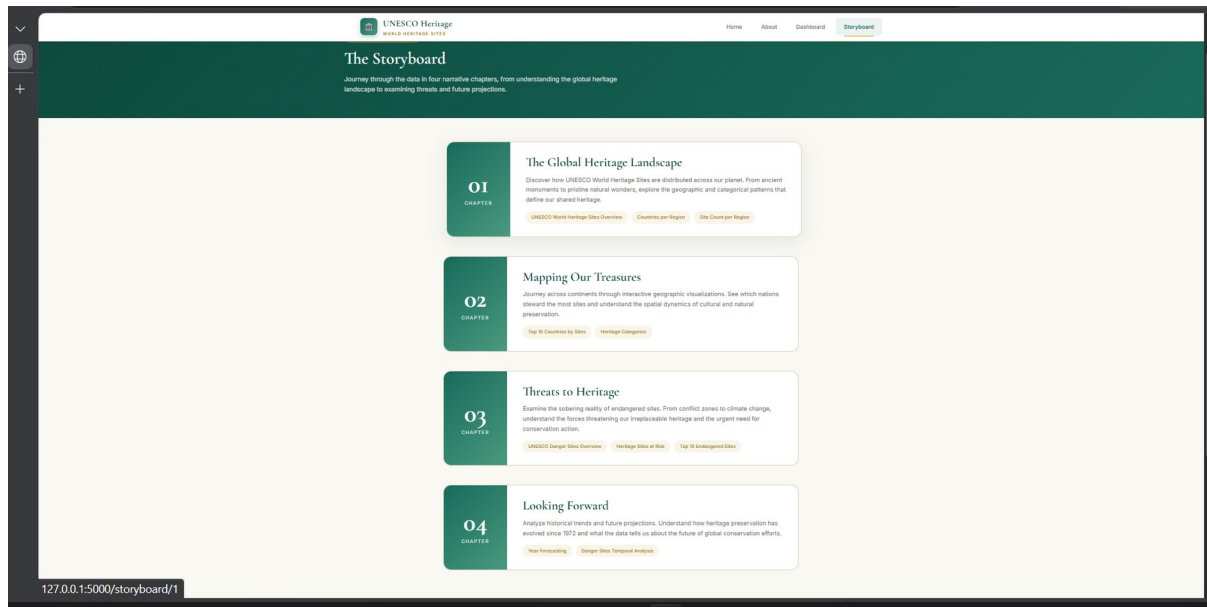
*Home page — project KPIs: 1,121 sites, 168 countries, 5 regions, 869 cultural, 213 natural, 53 endangered*



*About page — project background and objectives*



*Dashboard page — featured Tableau dashboards embedded in the site*



*Storyboard page — four-chapter narrative: Global Heritage Landscape, Mapping Our Treasures, Threats to Heritage, Looking Forward*

## 8. ADVANTAGES & DISADVANTAGES

### Advantages

- Free and openly accessible — no login required to view the dashboard.
- Highly interactive: filters, tooltips, and guided story navigation let viewers self-direct their exploration.
- Combines descriptive, geographic, and predictive views in one cohesive narrative rather than a single static chart.
- Built on a rigorously cleaned dataset with zero duplicate rows and zero HTML/encoding artifacts.
- The custom Flask front-end adds a polished, presentation-ready layer with KPIs and a narrative storyboard.
- Reusable pipeline (clean — connect — visualize — publish) that scales to future UNESCO updates or other open datasets.

### Disadvantages

- Requires an internet connection and a Tableau Public-compatible browser to view the live dashboard.
- Data is a static snapshot — it does not automatically refresh when UNESCO updates its official list.
- The 1,121-mark world map can take a few seconds to render on first load.
- Country/site labels can overlap at global zoom level, and there is no name-based search filter yet.
- No built-in mechanism for viewers to leave feedback or contact the project author.

## 9. CONCLUSION

The Heritage Treasures project successfully transformed a large, scattered, text-heavy UNESCO World Heritage dataset of 1,121 sites into an accessible, interactive analytics experience. Through careful data cleaning, thoughtful dashboard and story design in Tableau, and a purpose-built Flask website for presentation, the project delivers clear insight into how cultural and natural heritage is distributed globally, which sites are most at risk, and how the pace of heritage recognition has evolved since 1978. The result is a reusable, shareable resource for students, researchers, educators, and travelers alike.

## 10. FUTURE SCOPE

- Automate data refresh via UNESCO's official API or scheduled scraping, so the dashboard stays current as new sites are inscribed.
- Add a name/keyword search filter directly within the Tableau dashboard and the Flask website.
- Include representative photos for each heritage site to make the experience more visually engaging.
- Add multi-language support to broaden global accessibility.
- Explore predictive modeling of endangerment risk based on climate, conflict, and tourism-pressure indicators.
- Add a feedback/comments feature to the Flask website to gather viewer input.
- Package the Flask application as a mobile-responsive or native mobile experience.

## 11. APPENDIX

### Source Code (if any)

Flask web application & Tableau workbook source:

### Dataset Link

Official UNESCO World Heritage List: <https://whc.unesco.org/en/list/>

Cleaned working dataset (UNESCO\_Heritage\_Sites\_Cleaned.xlsx) is included with the project submission.

### GitHub & Project Demo Link

GitHub Repository:

UNESCO\_Heritage video link: [https://youtu.be/A2uFsZtX2mQ?si=\\_5Cip2lhoeWlamv9](https://youtu.be/A2uFsZtX2mQ?si=_5Cip2lhoeWlamv9)

Flask Project Demo: